



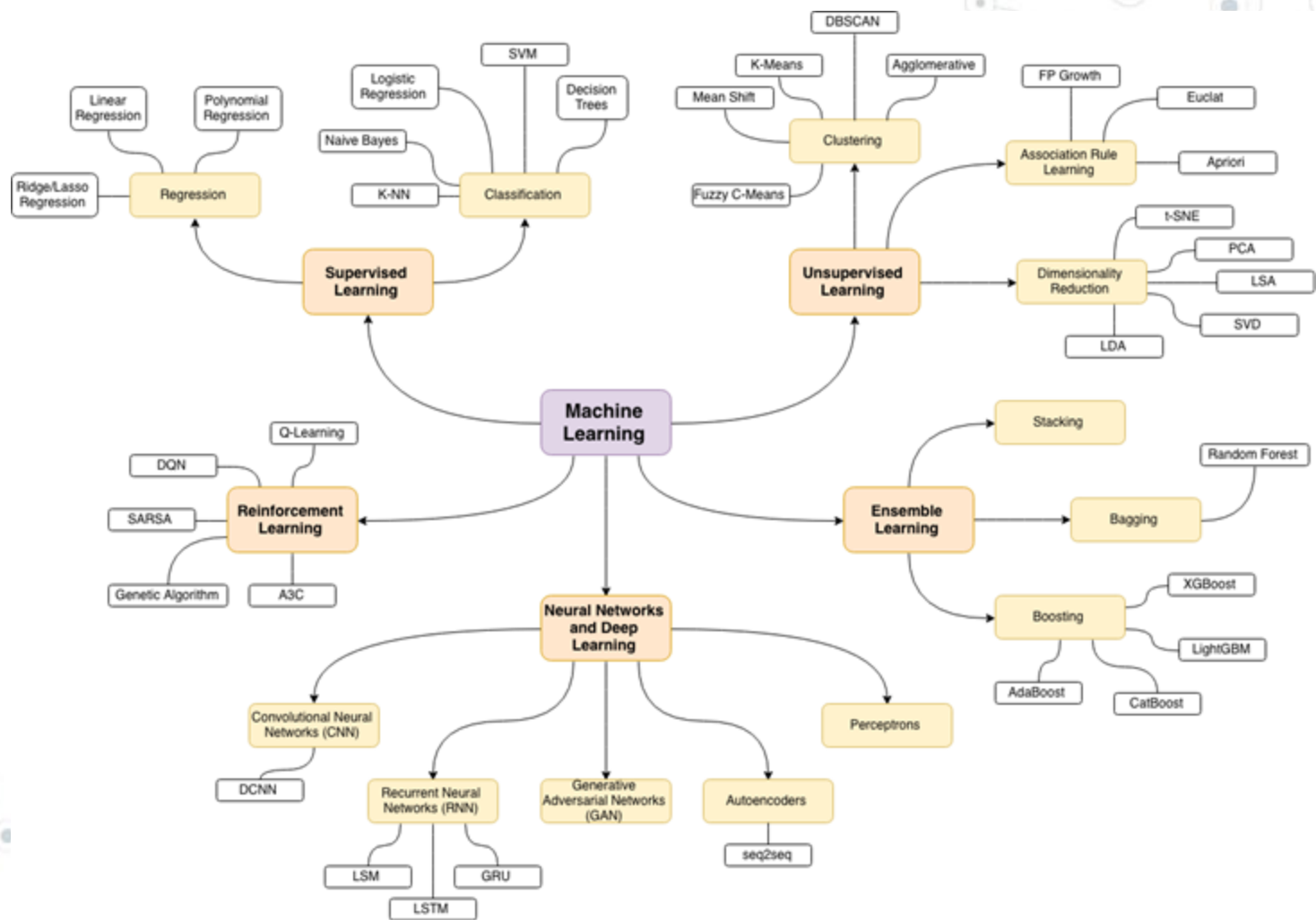
Lecture 1 - Introduction to ML

Presented By

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What is Machine Learning?

- Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from pre-trained data and generalize to unseen data, and thus perform tasks without being explicitly programmed.



Fig. 1: Traditional programming VS Machine Learning

What is Machine Learning?

Example: Celsius to Fahrenheit (Traditional)

1. The Concept

The Logic is Known:

In traditional programming, we (the humans) understand the relationship beforehand.

We explicitly program the mathematical formula:

$$F = C \times 1.8 + 32$$

The rules are hard-coded by the developer.

2. The Implementation

We write the rule:

```
def to_fahrenheit(c):  
    # The rule is explicit:  
    return c * 1.8 + 32  
# Pass input, get answer  
print(to_fahrenheit(10))  
# Output: 50
```

We pass input data (10), and the function returns the calculated answer (50).

What is Machine Learning?

Example: Celsius to Fahrenheit (Machine Learning)

2. Machine Learning Approach

The Logic is Unknown (initially):

Instead of writing the formula, we provide the computer with raw examples (Data Pairs).

The Training Data:

Input (C)	Output (F)
-40	-40
0	32
10	50
100	212

The Process:

1 We assume a generic model structure:

$$\hat{y} = \mathbf{w} \cdot x + \mathbf{b}$$

2 The algorithm iterates through the data to minimize error.

3 **Result:** It learns the parameters: $\mathbf{w} \approx 1.8$, $\mathbf{b} \approx 32$

Key Insight: In ML, the computer "figures out" the relationship (1.8 and 32) purely by analyzing the data, without being explicitly told the formula.

When ML is useful?

- When not all scenarios can be covered with if-else

Example: Image classification – a cat can look infinite ways; rules fail.

- When patterns exist but are hard to define manually

Example: Spam detector – spam emails don't follow fixed keywords; ML learns sneaky patterns.

- When sufficient data is available (labeled or unlabeled)

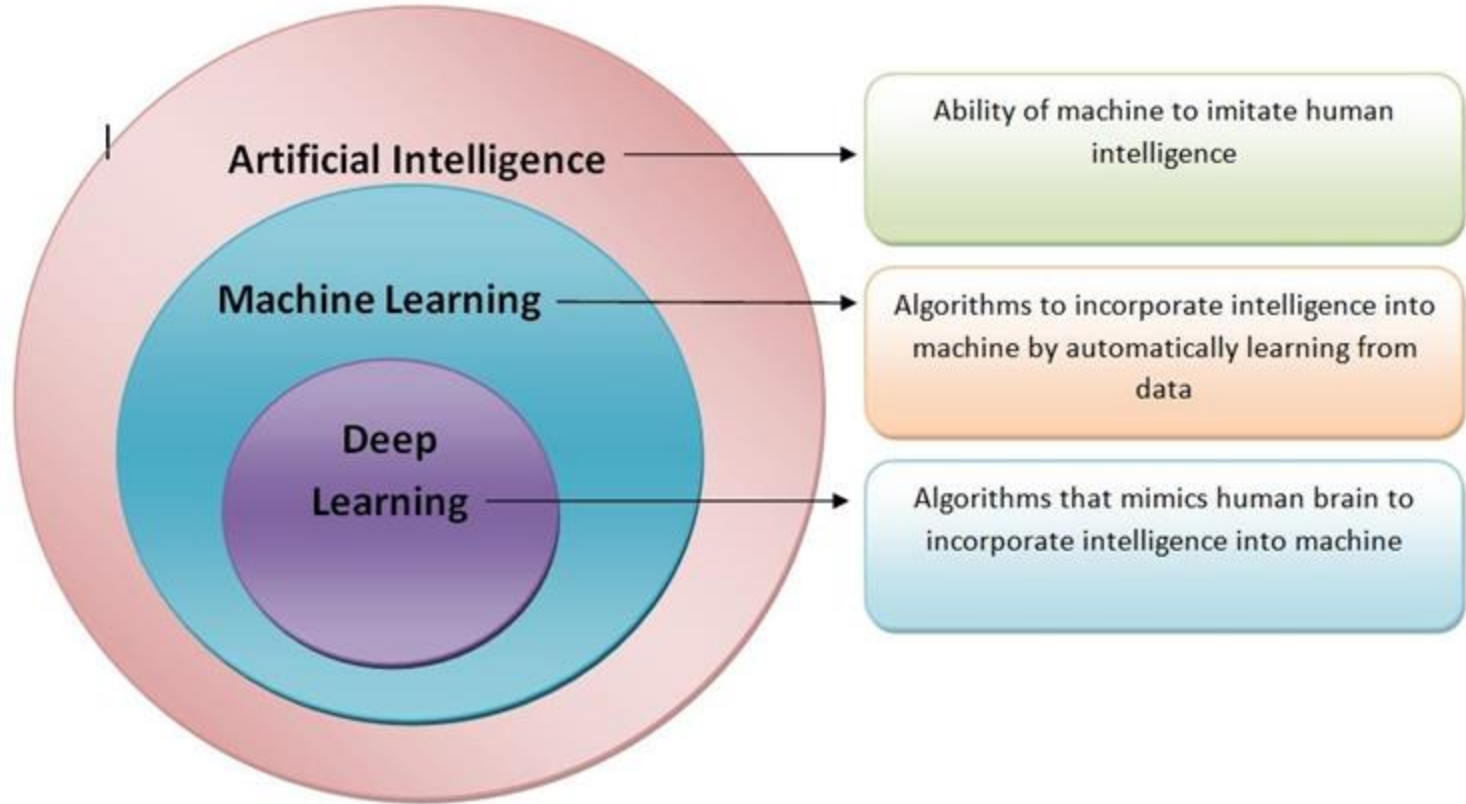
Example: Predicting house prices (labeled data) or customer grouping (unlabeled data)

Short History

Machine Learning isn't new. It started as a mathematical idea over 200 years ago, but practical use took off in the last 50+ years.

- 1795 – Legendre published the method of least squares (the math behind linear regression). This is the foundation of many ML algorithms. So ML's roots are ~230 years old, not just 50.
- 1950s–1980s – Early ML: perceptrons (simple neural networks), decision trees. But computers were slow and data was scarce.
- 1990s–2000s – Practical ML algorithms like SVM, random forests, and boosting emerged. Still limited by data and compute.
- Since ~2010 – Paradigm shift:
 - Explosion of devices (smartphones, sensors, IoT) and internet usage → massive amounts of digital data became available.
 - Improved hardware (GPUs, more RAM, SSDs) made training large models feasible.
 - Result: Deep learning took off — image recognition, voice assistants, recommendation systems became accurate and practical.

AI vs ML vs DL



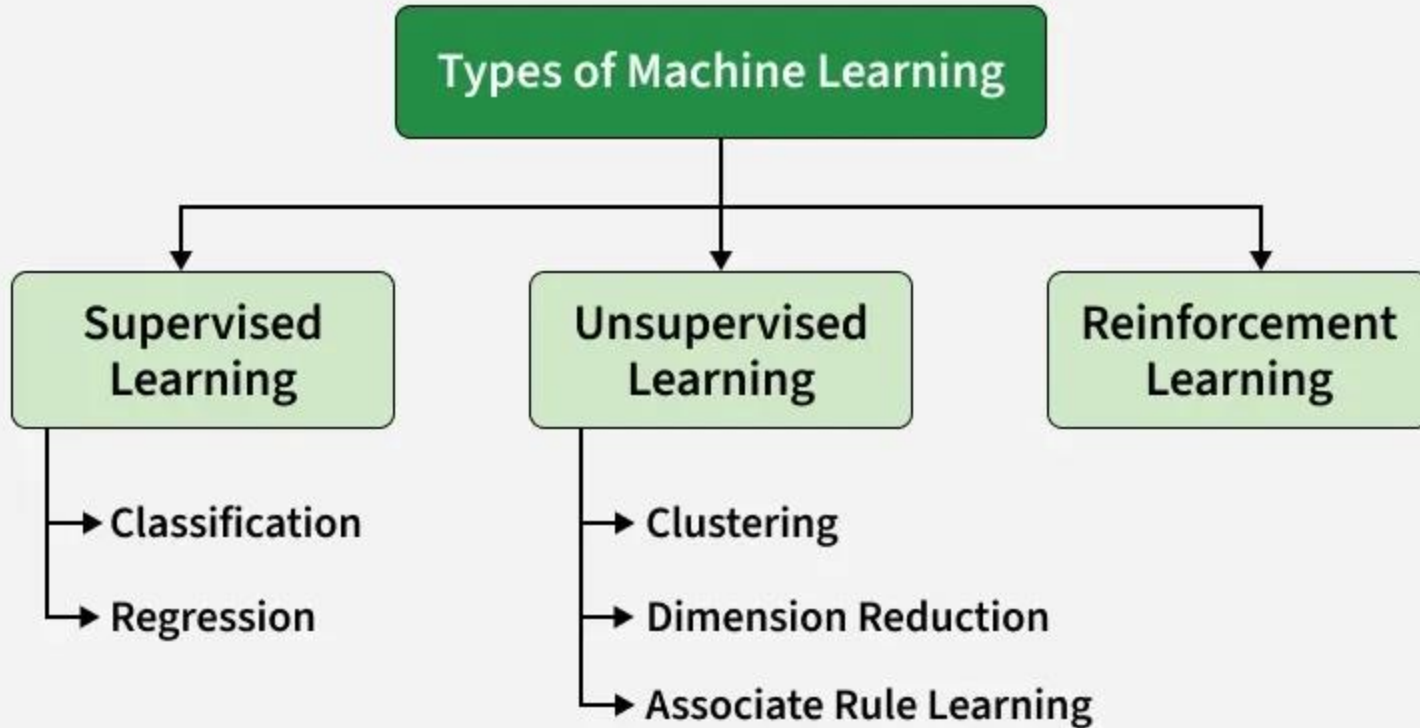
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5. TEST AND VALIDATE MODEL

6. DEPLOY MODEL



Types of ML



Supervised ML

Definition: Learning a mapping function f from input variables x to output variables y based on example input-output pairs.

- **Goal:** Approximate $y = f(x)$ well enough to predict y for new x .
- **Dataset:** $D = \{(x^{(i)}, y^{(i)})\}_{i=1}^m$

Regression

Predicting continuous values.

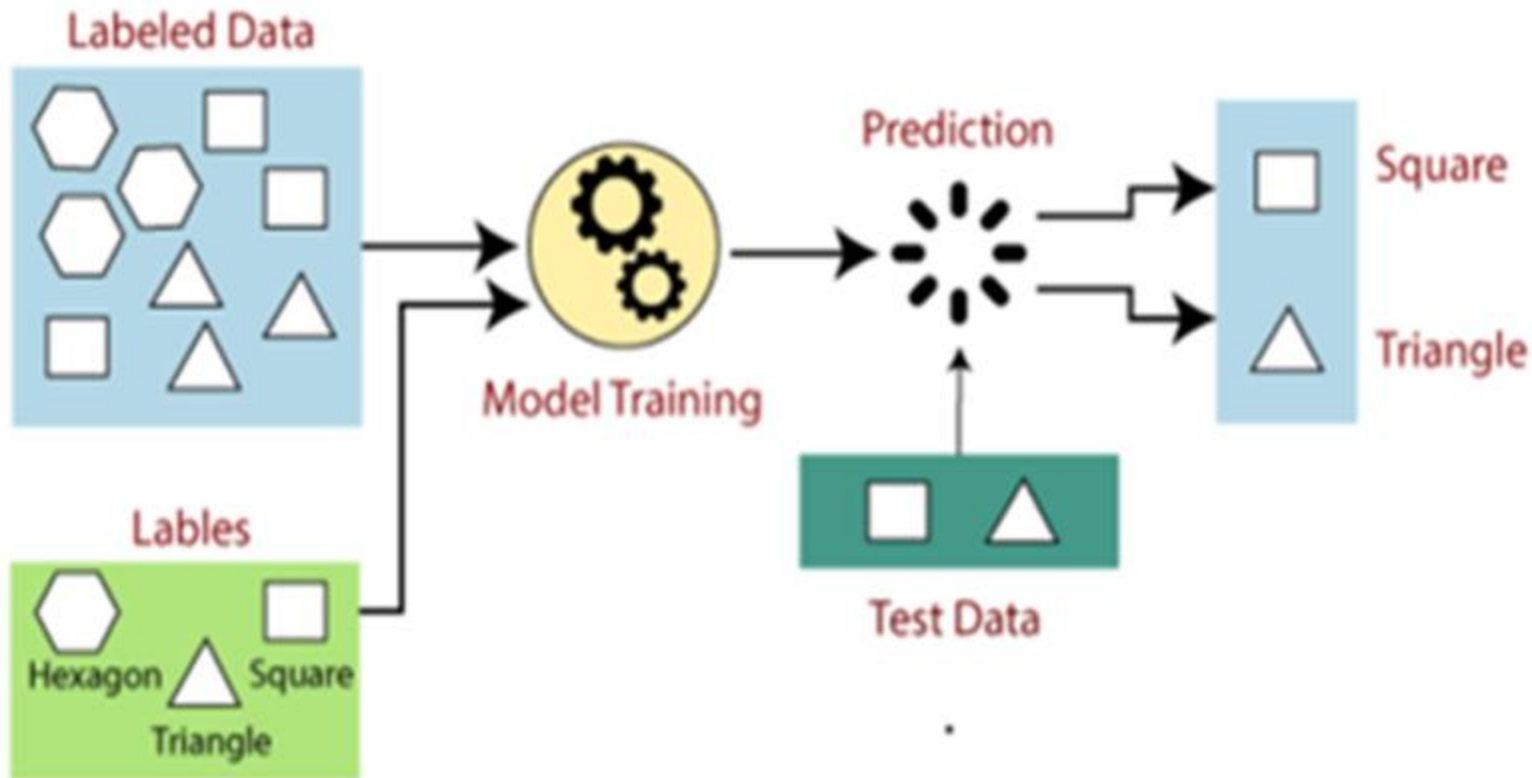
- *Ex:* House prices, Temperature.
- $y \in \mathbb{R}$

Classification

Predicting discrete labels.

- *Ex:* Spam detection, Diagnostics.
- $y \in \{0, 1, \dots, K\}$

Supervised ML



Applications of Supervised Learning

1. Classification (Discrete Output)

Predicting a category or class label.

- **Image Recognition:** Face detection, identifying objects in photos (e.g., Google Photos).
- **Spam Filtering:** Classifying emails as "Spam" or "Not Spam" (e.g., Gmail).
- **Medical Diagnosis:** Predicting if a tumor is malignant or benign based on scans.
- **Fraud Detection:** Flagging credit card transactions as fraudulent or legitimate.

2. Regression (Continuous Output)

Predicting a numerical value.

- **Real Estate Prediction:** Estimating house prices based on square footage, location, and rooms.
- **Stock Market Forecasting:** Predicting future stock prices based on historical trends.
- **Weather Forecasting:** Predicting temperature, humidity, or rainfall amount.
- **Demand Forecasting:** Estimating product sales for retail inventory planning.

Supervised Learning is the most widely used form of ML in industry today.

Unsupervised ML

Definition: Inferred function to describe hidden structure from "unlabeled" data.

- **Data:** $D = \{x^{(i)}\}_{i=1}^m$ (No labels y).
- **Goal:** Find patterns, groupings, or compressed representations.

Main Types:

- 1 **Clustering:** Grouping similar data points together (e.g., k-Means).

$$\min_{c, \mu} \sum_{i=1}^m \|x^{(i)} - \mu_{c(i)}\|^2$$

- 2 **Dimensionality Reduction:** Reducing the number of random variables under consideration (e.g., PCA).
- 3 **Association:** Discovery of rules (e.g., Market Basket Analysis).

Applications of Unsupervised Learning

1. Clustering (Grouping)

Grouping similar data points together.

- **Customer Segmentation:** Grouping customers by purchasing behavior for targeted marketing (e.g., "Budget Shoppers" vs. "High Spenders").
- **Anomaly Detection:** identifying unusual patterns in network traffic (Cybersecurity) or manufacturing (Fault detection).
- **Social Network Analysis:** Detecting communities or circles of friends.

2. Association & Dim. Reduction

Finding rules and simplifying data.

- **Market Basket Analysis:** Discovering items frequently bought together (e.g., "People who buy Bread also buy Butter").
- **Recommendation Systems:** Suggesting movies/products based on user similarity (e.g., Netflix, Amazon).
- **Feature Compression:** Reducing file size or complexity without losing information.

Unsupervised Learning excels at discovering hidden structures in unlabeled data.

Reinforcement Learning

Definition

Learning **what to do how** to map situations to actions to maximize a numerical **reward signal**.

Key Components:

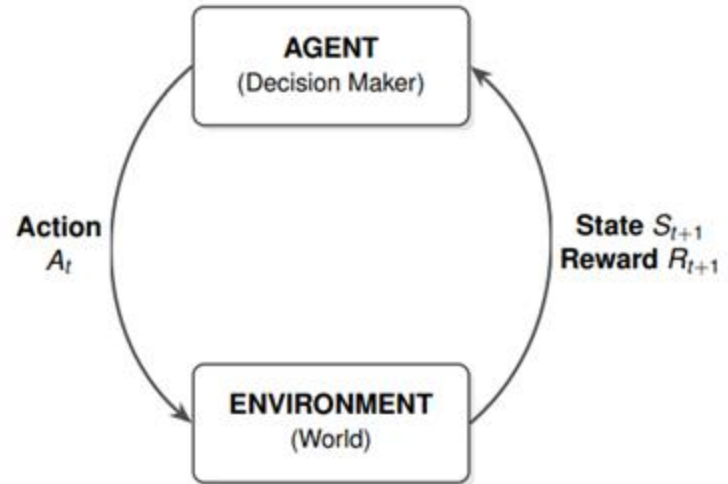
Agent: The learner & decision maker.

Environment: Everything the agent interacts with.

Action (A_t): What the agent chooses to do.

State (S_t): The current situation/context.

Reward (R_t): Immediate feedback score (Good/Bad).



Goal: Maximize the total cumulative reward over time.

Applications of Reinforcement Learning

1. Robotics & Control

Interacting with the physical world.

- **Autonomous Vehicles:** Learning to navigate traffic, merge lanes, and park without crashing.
- **Industrial Robotics:** Arms learning to pick and place objects of varying shapes and sizes.
- **Data Center Cooling:** Google's DeepMind reduced cooling costs by 40% via RL.

2. Games & Strategy

Beating humans in complex environments.

- **Game Playing (AlphaGo):** Mastering Go, Chess, and StarCraft II by playing millions of games against itself.
- **Recommendation Systems:** Optimizing long-term user engagement (e.g., YouTube next video suggestion).
- **Personalized Education:** Tutors that adapt lesson difficulty based on student performance.

RL learns complex behaviors through trial and error to maximize rewards.